TASK 1: PACKET IDENTIFICATION USING WIRESHARK

Roll Number: 2520030443

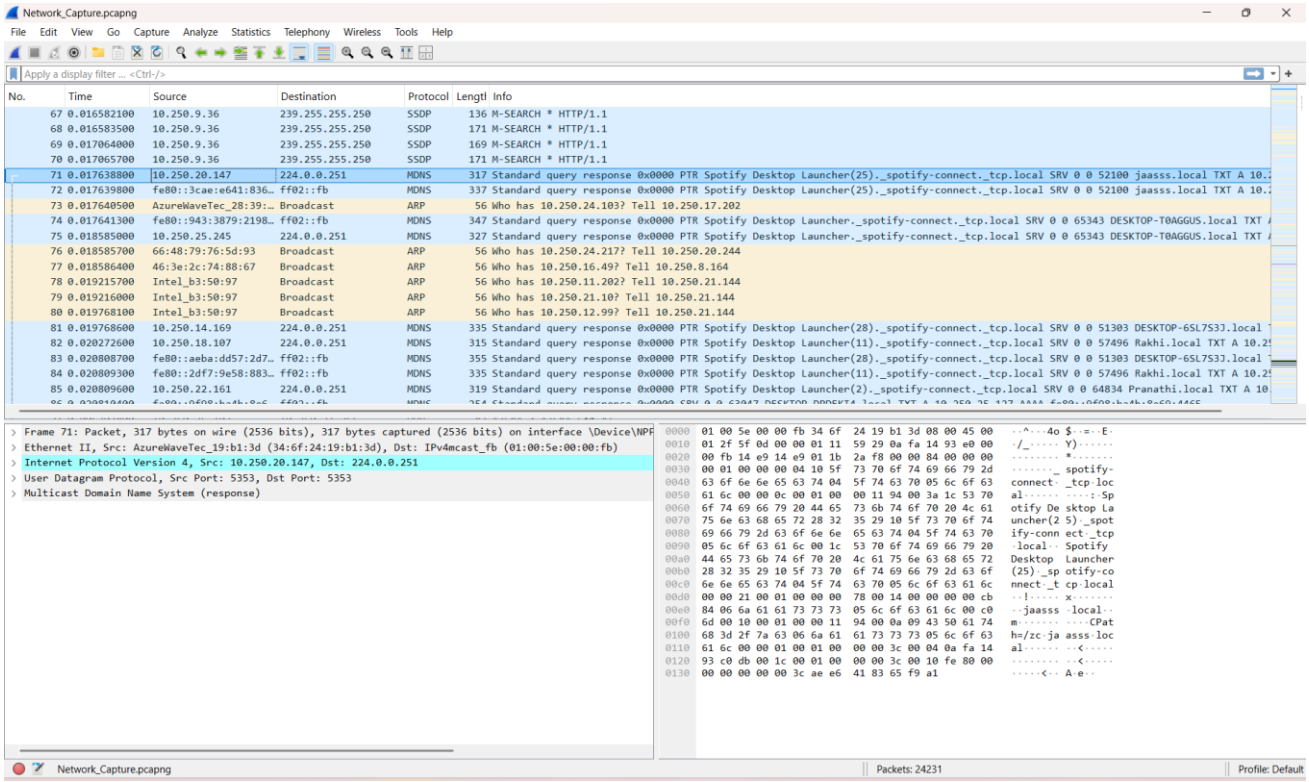
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1. Identify the Total Number of Packets Captured

Packet: A packet is a small unit of data transmitted over a computer network.

Procedure:

1. Open the Wireshark application on the computer.
2. Click on File → Open and select the previously created PCAP/PCAPNG file.
3. Click Open to load the captured network packets.
4. Wait until all the packets are displayed in the packet list.
5. Observe the bottom status bar of the Wireshark window.
6. Note the total number of packets captured.



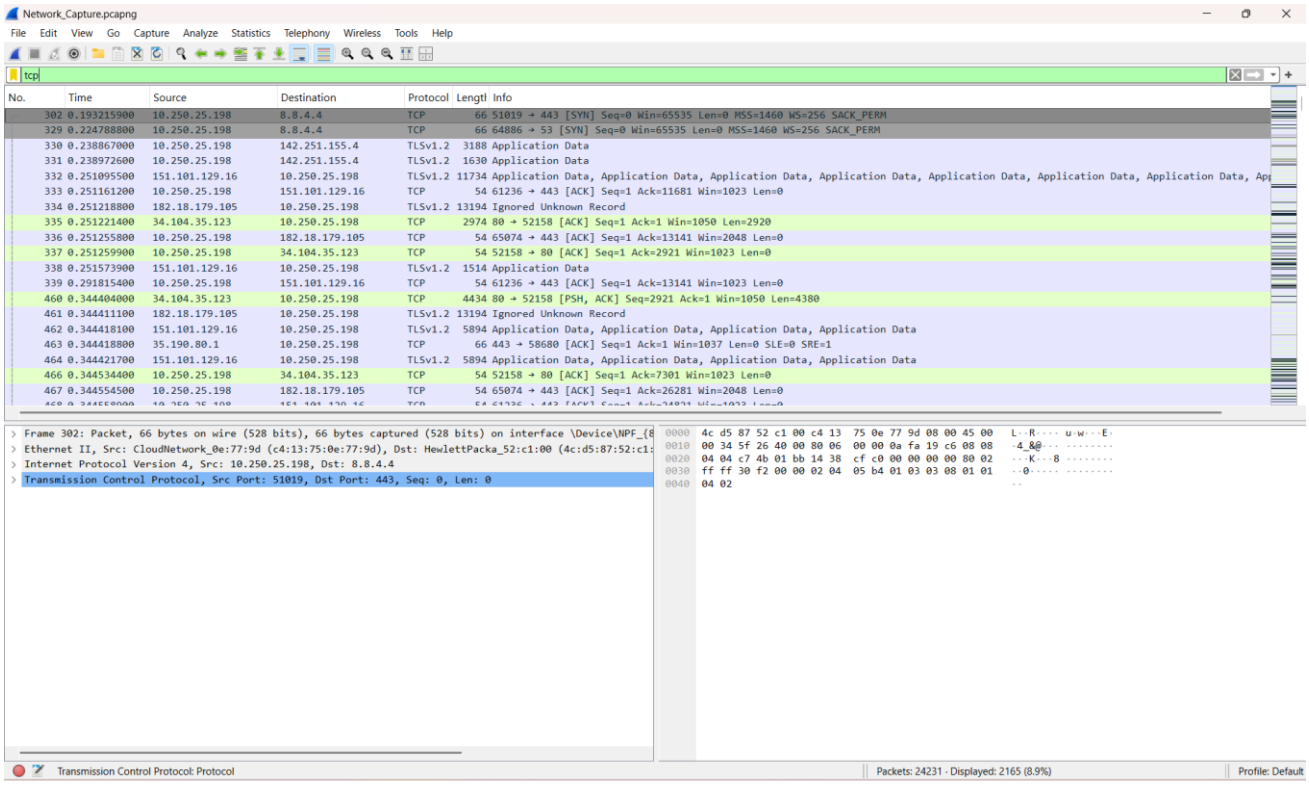
Result: The total number of packets captured was 24231.

2. Identify the Number of TCP Packets

TCP: TCP is a connection-oriented protocol that provides reliable data transmission between devices.

Procedure:

1. Open the PCAP file in Wireshark.
2. Locate the display filter bar at the top of the window.
3. Enter the filter tcp in the display filter bar.
4. Press the Enter key to apply the filter.
5. Wireshark displays only the TCP packets.
6. Observe the number of displayed packets at the bottom of the window.
7. Record the TCP packet count.



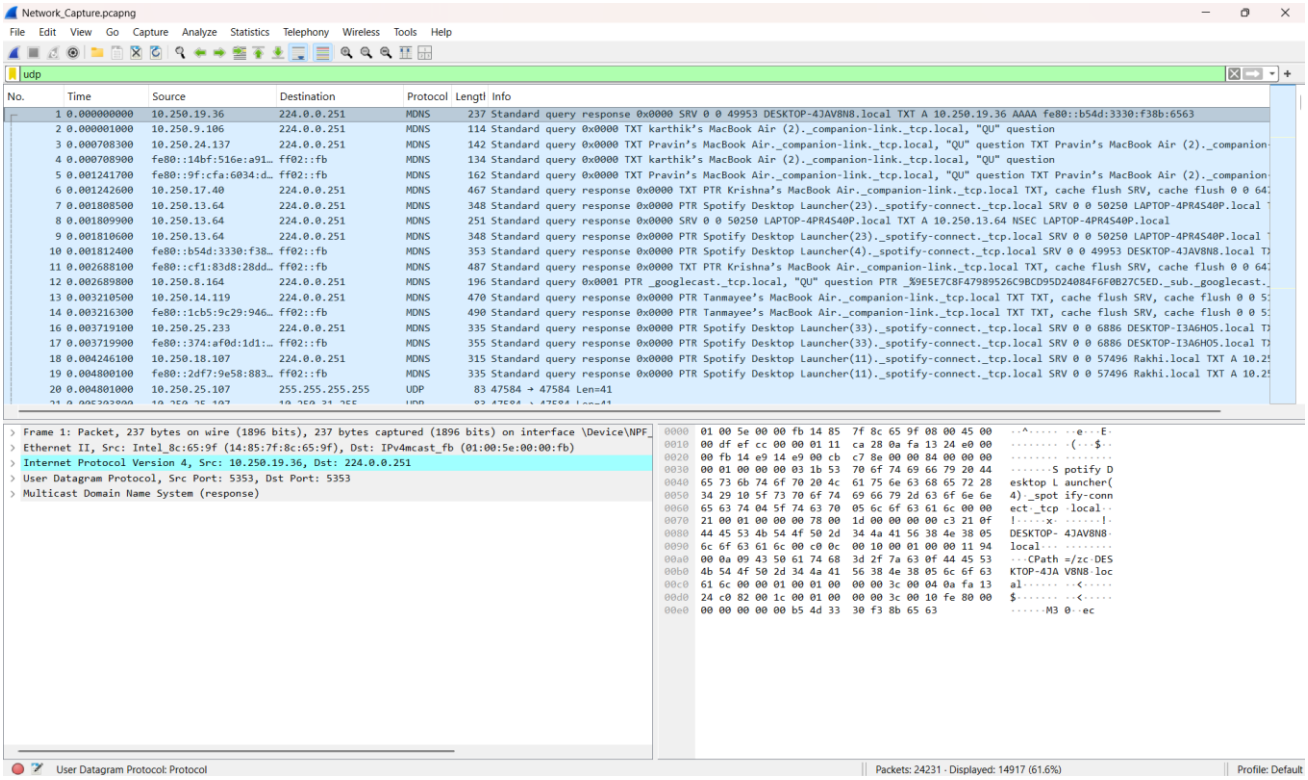
Result: The total number of TCP packets was 2165.

3. Identify the Number of UDP Packets

UDP: UDP is a connectionless protocol that sends data quickly without guaranteeing delivery.

Procedure:

1. Open the same PCAP file in Wireshark.
2. Clear the previous display filter.
3. Enter the filter udp in the display filter bar.
4. Press Enter to apply the filter.
5. Wireshark displays only the UDP packets.
6. Observe the displayed packet count at the bottom of the window.
7. Record the UDP packet count.



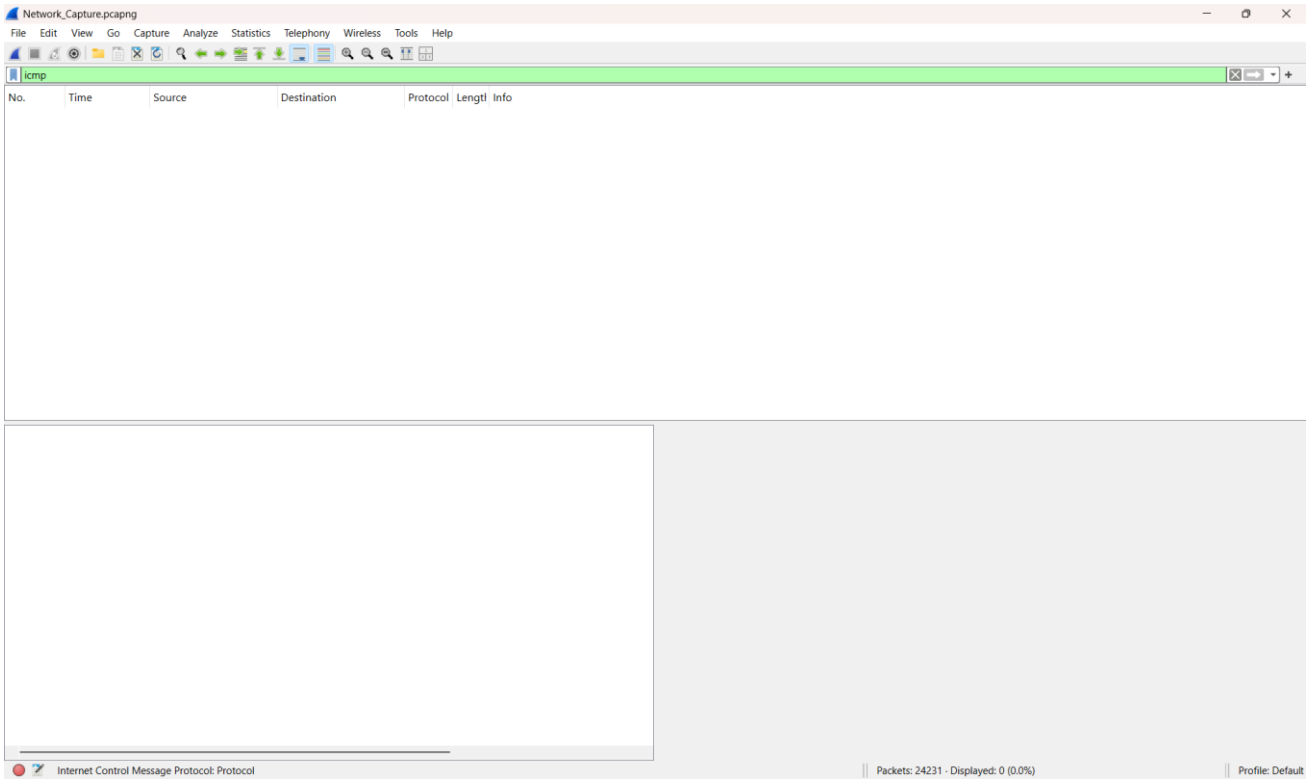
Result: The total number of UDP packets was 14917.

4. Identify the Number of ICMP Packets

ICMP: ICMP is a network protocol used to send error messages and diagnostic information. It is commonly used by the ping command.

Procedure:

1. Open the same PCAP file in Wireshark.
2. Clear the previous display filter.
3. Enter the filter icmp in the display filter bar.
4. Press Enter to apply the filter.
5. Wireshark displays only the ICMP packets.
6. Observe the number of displayed packets at the bottom of the window.
7. Record the ICMP packet count.



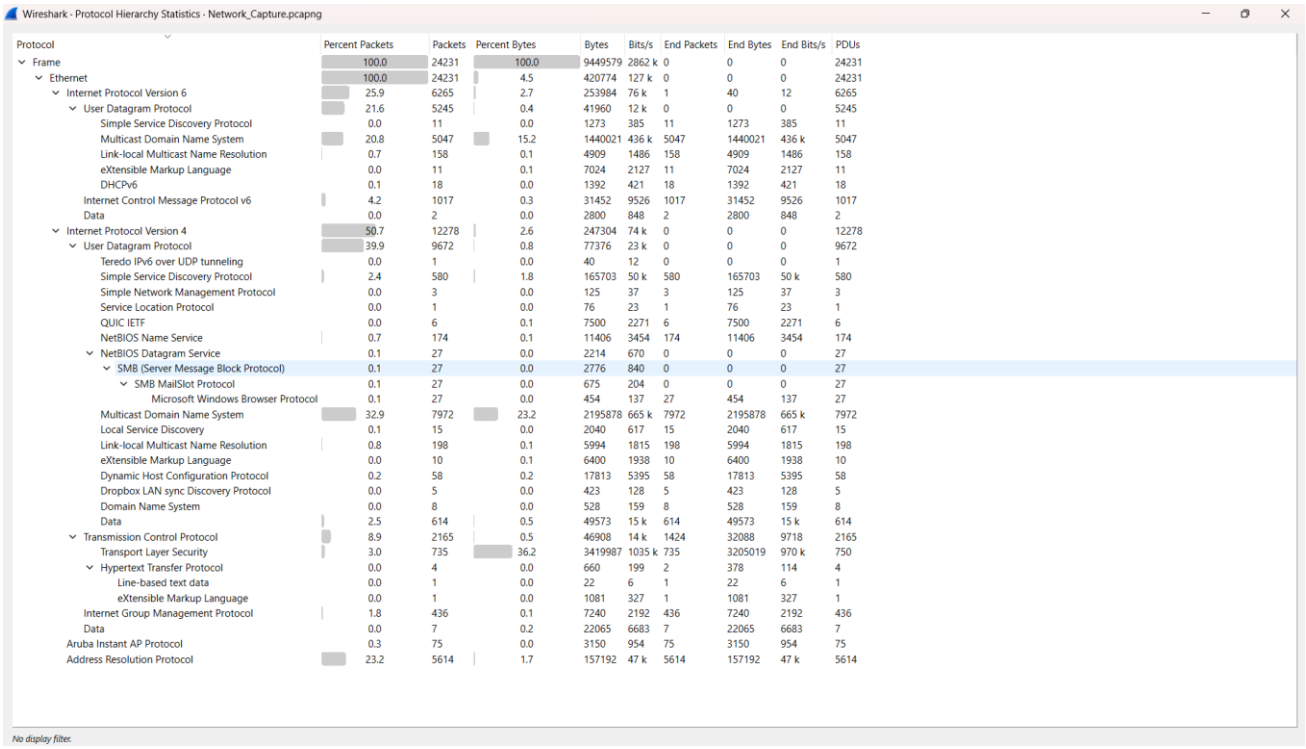
Result: No ICMP packets were found in the captured network traffic. The displayed packet count was 0.

5. Identify the Top Five Protocols Using Protocol Hierarchy

Protocol Hierarchy: Protocol Hierarchy is a Wireshark statistics feature that displays the protocols present in a capture along with their packet counts and percentages.

Procedure:

1. Open the PCAP file in Wireshark.
2. Remove any previously applied display filters.
3. Click on the Statistics menu in the menu bar.
4. Select Protocol Hierarchy from the dropdown menu.
5. A Protocol Hierarchy window will appear.
6. Observe the list of protocols displayed in the window.
7. Identify the top five protocols based on their packet counts.
8. Record the protocol names and their corresponding packet counts.



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	24231	100.0	9449579	2862 k	0	0	0	24231
Ethernet	100.0	24231	4.5	420774	127 k	0	0	0	24231
Internet Protocol Version 6	25.9	6265	2.7	253984	76 k	1	40	12	6265
User Datagram Protocol	21.6	5245	0.4	41960	12 k	0	0	0	5245
Simple Service Discovery Protocol	0.0	11	0.0	1273	385	11	1273	385	11
Multicast Domain Name System	20.8	5047	15.2	1440021	436 k	5047	1440021	436 k	5047
Link-local Multicast Name Resolution	0.7	158	0.1	4909	1486	158	4909	1486	158
eXtensible Markup Language	0.0	11	0.1	7024	2127	11	7024	2127	11
DHCPv6	0.1	18	0.0	1392	421	18	1392	421	18
Internet Control Message Protocol v6	4.2	1017	0.3	31452	9526	1017	31452	9526	1017
Data	0.0	2	0.0	2800	848	2	2800	848	2
Internet Protocol Version 4	50.7	12278	2.6	247304	74 k	0	0	0	12278
User Datagram Protocol	39.9	9672	0.8	77376	23 k	0	0	0	9672
Teredo IPv6 over UDP tunneling	0.0	1	0.0	40	12	0	0	0	1
Simple Service Discovery Protocol	2.4	580	1.8	165703	50 k	580	165703	50 k	580
Simple Network Management Protocol	0.0	3	0.0	125	37	3	125	37	3
Service Location Protocol	0.0	1	0.0	76	23	1	76	23	1
QUIC IETF	0.0	6	0.1	7500	2271	6	7500	2271	6
NetBIOS Name Service	0.7	174	0.1	11406	3454	174	11406	3454	174
NetBIOS Datagram Service	0.1	27	0.0	2214	670	0	0	0	27
SMB (Server Message Block Protocol)	0.1	27	0.0	2776	840	0	0	0	27
SMB MailSlot Protocol	0.1	27	0.0	675	204	0	0	0	27
Microsoft Windows Browser Protocol	0.1	27	0.0	454	137	27	454	137	27
Multicast Domain Name System	32.9	7972	23.2	2195878	665 k	7972	2195878	665 k	7972
Local Service Discovery	0.1	15	0.0	2040	617	15	2040	617	15
Link-local Multicast Name Resolution	0.8	198	0.1	5994	1815	198	5994	1815	198
eXtensible Markup Language	0.0	10	0.1	6400	1938	10	6400	1938	10
Dynamic Host Configuration Protocol	0.2	58	0.2	17813	5395	58	17813	5395	58
Dropbox LAN sync Discovery Protocol	0.0	5	0.0	423	128	5	423	128	5
Domain Name System	0.0	8	0.0	528	159	8	528	159	8
Data	2.5	614	0.5	49573	15 k	614	49573	15 k	614
Transmission Control Protocol	8.9	2165	0.5	46908	14 k	1424	32088	9718	2165
Transport Layer Security	3.0	735	36.2	3419987	1035 k	735	3205019	970 k	750
Hypertext Transfer Protocol	0.0	4	0.0	660	199	2	378	114	4
Line-based text data	0.0	1	0.0	22	6	1	22	6	1
eXtensible Markup Language	0.0	1	0.0	1081	327	1	1081	327	1
Internet Group Management Protocol	1.8	436	0.1	7240	2192	436	7240	2192	436
Data	0.0	7	0.2	22065	6683	7	22065	6683	7
Aruba Instant AP Protocol	0.3	75	0.0	3150	954	75	3150	954	75
Address Resolution Protocol	23.2	5614	1.7	157192	47 k	5614	157192	47 k	5614

Overall Conclusion

Wireshark was used to analyze the captured network traffic. The total number of packets was identified, and TCP, UDP, and ICMP packets were filtered using display filters. The Protocol Hierarchy feature was used to identify the top five protocols and understand the distribution of network traffic.